Holden Anderson

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EDUCATION

California State University, San Bernardino — San Bernardino, CA
Bachelor of Science in Computer Science — *Aug. 2023 – May 2027*
Coursework: Data Structures & Algorithms, Databases, Machine Learning, Statistics, Linear Algebra

EXPERIENCE

Co-Founder & Engineer — Nov. 2025 – Present
Lovely LLC | Lovely: Relationship Growth — iOS App — *Remote*

- Built the FastAPI backend for an AI relationship-coaching app live on the App Store, routing requests across OpenAI and Anthropic models with per-user conversation memory
- Protected user data with per-user encryption, rate limiting, and signed webhook verification
- Wrote default-deny Firestore security rules that isolate each user's data, covered by 30+ pytest test files
- Built subscription billing and free trials with RevenueCat, gaining the app's first paying subscribers after launch

Lead Engineer — Apr. 2026 – Sept. 2026
Patch | P2P Payment Platform — *Remote*

- Led a 2-engineer team building a peer-to-peer payments app (React Native, FastAPI) with 3 payment methods: offline codes, ultrasonic audio, and real-time phone-to-phone matching over WebSockets
- Signed and verified ultrasonic audio payments with HMAC-SHA256 in under 2 seconds, rejecting tampered payments on-device before they reached the server
- Used Redis pub/sub to deliver real-time WebSocket events across multiple server instances on AWS ECS Fargate
- Secured payment PINs with bcrypt hashing and voided payments after 5 failed attempts

PROJECTS

NFL Game Predictor | *Python, pandas, scikit-learn, Claude Code* | github — 2026

- Built an ML model that predicts NFL game winners with 64.3% accuracy on 544 held-out games (vs. 54.4% baseline), using 67 engineered features including custom Elo ratings
- Prevented data leakage with season-by-season walk-forward validation and automated leak checks; held 61.6% accuracy on an untouched 2025 test season
- Benchmarked against Vegas odds (69.7%) and deployed the model for live weekly 2026 predictions

swapLLM & swaptrace | *Python, Pydantic, pytest* | github | swapllm | swaptrace — 2026

- Published a Python package to PyPI that sends requests to Groq, OpenAI, or Anthropic through one API, automatically falling back to the next provider on rate limits, timeouts, or outages
- Wrote a 96-test pytest suite using mocked API responses, plus optional Pydantic validation of model output
- Published swaptrace, a companion PyPI package that records every fallback attempt and compares provider reliability and latency from the command line

Kache | *Python, TCP Sockets, Docker, AWS EC2* | github — 2026

- Built a Redis-like key-value cache from raw TCP sockets in Python with no dependencies, sustaining 17,000+ ops/sec at under 2ms median latency across 20 concurrent clients
- Added primary/replica replication and append-only file persistence; replicas fully resynced in under 2 seconds after a simulated network outage with zero lost writes

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Frameworks & Libraries: FastAPI, React Native (Expo), React, SQLAlchemy, Pydantic, pytest, pandas, scikit-learn
Cloud & Tools: AWS (EC2, ECS Fargate, ElastiCache), Docker, PostgreSQL, Redis, Firebase, Git, GitHub Actions
AI: OpenAI API, Anthropic API, Groq, Claude Code
Concepts: REST APIs, WebSockets, JWT auth, CI/CD, unit & integration testing, LLM prompt orchestration